

PPG Tau PET (AV-1451) Methods: Processing and Quality Control

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Overview

The following document describes the methods for processing tau-PET data using [^{18}F] AV-1451 flortaucipir (FTP). These procedures are consistent with the FTP protocol provided by the Jagust Lab at UC Berkeley for ADNI (revised, 01/14/2021). All tau-PET scans were acquired on a Siemens Biograph64 TruePoint system (University of Southern California).

Key Points

- Regions of interest (ROIs) were defined for Braak stages I, III-IV, and V-VI, along with a meta-temporal composite. The inferior cerebellum was designated as the reference region and was extracted through the SUIT toolbox.
- The Advanced Normalization Toolbox (ANTs) was utilized to register each ROI and the reference region to PET space for the purpose of SUVR analysis.
- SUVRs were calculated by dividing the mean uptake within each region by the mean uptake in the inferior cerebellum. Both unweighted and volume-weighted SUVRs were computed for all composite ROIs.

Methods

Software and Tools:

- dcm2niiX (version v1.0.20240202)
- FreeSurfer (version 7.1.1)
- MATLAB R2019b
- SUIT (spatially unbiased atlas template of the cerebellum and brainstem)
- ANTS (Advanced Normalization Tools)
- FSL (FMRIB Software Library)

Processing Steps:

- Freesurfer Processing
- PET Preprocessing
 - Motion Correction
 - Averaging of Timeframes
- PET-MRI Registration
- SUIT Processing
- Calculation of SUVRs for Predefined Meta-ROIs
 - Non-weighted results
 - Volume-weighted results

Quality Control Steps

Internal QC codes to be used during processing

The following internal quality control (QC) codes are used to evaluate the quality of PET imaging data throughout the processing pipeline. Each scan is assessed based on specific criteria, and the QC code is assigned based on the overall suitability of the scan for further analysis. These codes help streamline the identification of scans that are ready for analysis, require correction, or should be excluded due to quality issues.

QC_RESULT	DESCRIPTION
External Use	
Pass	The scan meets all quality control criteria and is suitable for analysis.
Partial Pass	The data is in an acceptable format. However, minor issues are present. These may include slight under-segmentation of ROIs in ~1/2 slices or minor motion in the PET image.
Fail	The scan is excluded from analysis due to persistent issues after standard troubleshooting - such as severe motion artifacts, Freesurfer parcellation issues due to pathology, or consistent registration failure.
Internal Use	
Requires Fixing	The scan has significant issues such as heavy under-segmentation of ROIs/reference region, severe motion artifacts, or major PET-MR registration errors. These steps require troubleshooting and reprocessing.

Step 1: FreeSurfer Processing

- Accelerated sagittal MPRAGE (3T) scans are acquired from the LONI-IDA repository and converted from DICOM to NIFTI format using dcm2niix.
- Raw T1-weighted scans are passed into FreeSurfer's recon-all pipeline. The resulting Desikan-Killiany (aparc+aseg) (Figure 1) and Destrieux (aparc.a2009s+aseg) atlases are utilized for further processing.

QC Process: The atlas is overlaid on the T1 MRI and checked for segmentation issues.

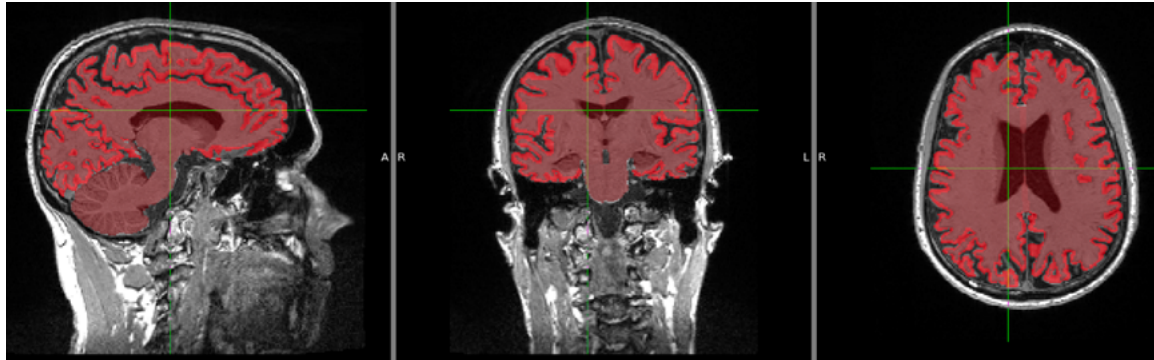


Figure 1: The figure shows the FreeSurfer aparc+aseg mask in T1 space overlaid on the respective T1 MRI.

Step 2: PET Pre-processing

- Tau-PET scans are downloaded from the LONI-IDA repository and converted from DICOM to NIFTI format using dcm2niix. PET scans collected within 8 months of the respective T1-weighted MRI are selected for further processing.
- **Motion Correction:** The PET image is smoothed using a 6mm full-width half-maximum (FWHM) Gaussian kernel. After smoothing, motion correction is performed on the smoothed image using FSL's mcflirt tool, employing normalized mutual information (normmi) as the cost function to optimize alignment between frames. The calculated motion correction transformations are then applied to the original, unsmoothed PET image.
- **Averaging of Timeframes:** Following motion correction, the individual frames of the PET scan are averaged.

QC Process for Motion Correction:

- The motion-corrected PET image is overlaid on the raw PET scan. We scroll through each volume and compare them.
- If significant pixelation or distortion around the head occurs in multiple volumes, this step is re-run.

- fsl_motion_outliers is applied to the motion-corrected PET image to identify outlier frames based on framewise displacement (FD) values exceeding a threshold of 0.9 millimeters.

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Step 3: PET-MRI Registration

- The motion-corrected mean PET image is registered to anatomical space using the ANTs registration tool. This process starts with bias correction of the T1 scan via ANTs N4 Bias Correction, followed by skull stripping using FreeSurfer's mri_watershed algorithm.
- The PET image is linearly registered to the N4-corrected T1, followed by a second round of rigid registration using the skull-stripped brain as the reference.
- The transformation matrices obtained from these steps are applied to the motion-corrected mean PET image, which is subsequently smoothed with an 8 mm Gaussian kernel.

QC of PET-MRI Registration:

- The smoothed PET image is overlaid on the T1 scan. If the skull, meninges, cortex, and ventricles align well with the native T1 image, the subject is passed (Figure 2).

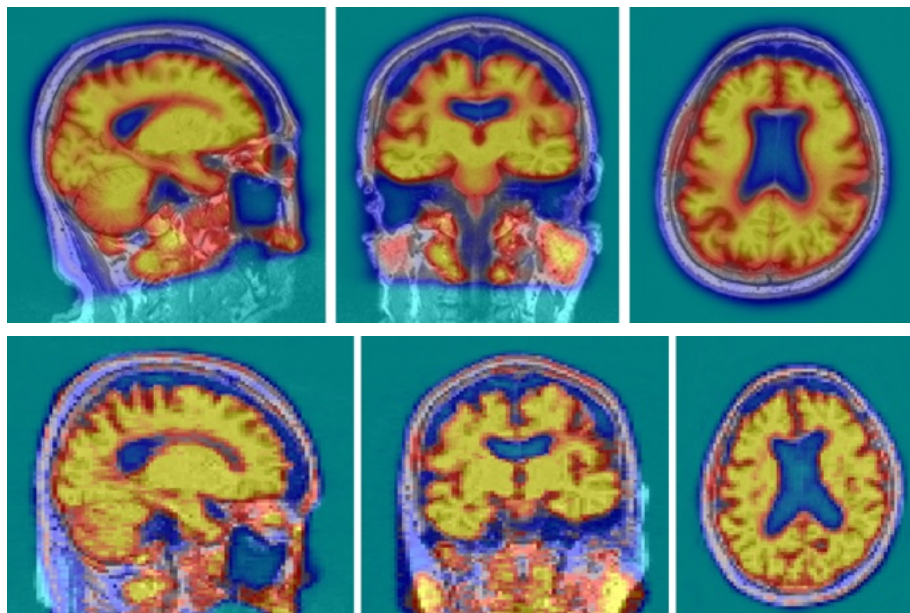


Figure 2: The top panel displays the PET image in T1 space, while the lower panel displays the inverse (T1 in PET space). Both registrations are checked prior to warping the ROIs and reference region to PET space.

Step 4: Extraction of the Reference Region and ROI Masks

SUIT Processing

- For FTP analysis, the inferior cerebellum is designated as the reference region, extracted by processing the native T1 MRI through the SUIT toolbox (spatially unbiased atlas template of the cerebellum and brainstem; <http://www.diedrichsenlab.org/imaging/suit.htm>).
- Through the application of suit toolbox functions (`suit_isolate_seg`, `suit_normalize`), the cerebellum is isolated from the T1-weighted image and resliced to SUIT space (Figure 3).



Figure 3: The figure shows the T1 extracted cerebellum aligned to SUIT space.

- These steps result in the production of a deformation field which is utilized to transform the cerebellar atlas from SUIT to T1 space (`suit_reslice_inverse`) (Figure 4). ANTs is applied to enhance this transformation.

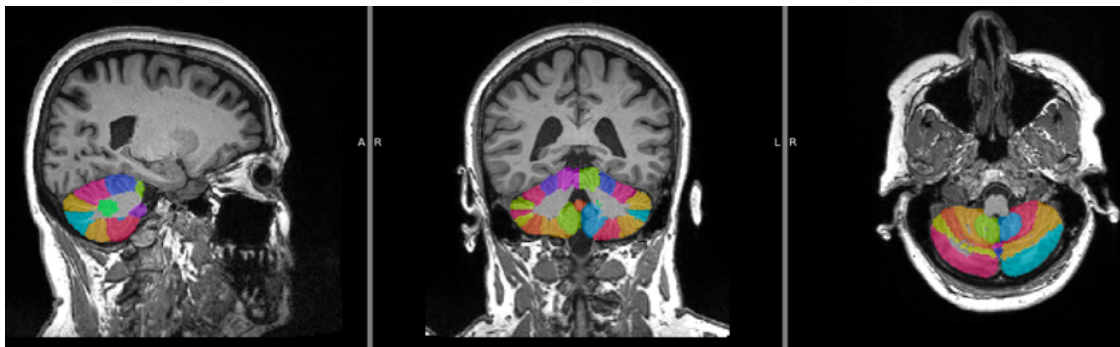


Figure 4: The figure shows the SUIT cerebellar atlas aligned with the subject's T1 MRI.

- Once the SUIT atlas is aligned to the subject's T1 MRI, the atlas is thresholded to extract regions corresponding to the superior (SUIT codes: 1-5, 7) and inferior cerebellum (codes: 6, 8-28, 33, 34). The inferior cerebellum is crossed with a Freesurfer derived cerebellar gray matter mask and refined by subtracting out dorsal regions (Figure 5).

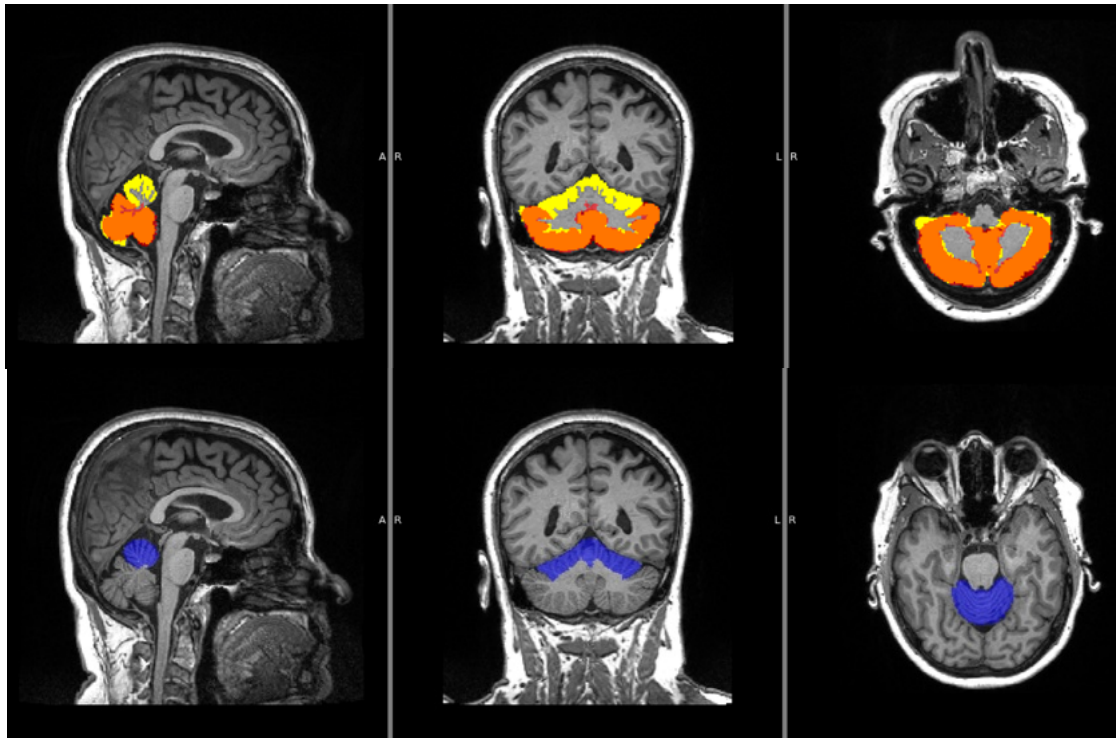


Figure 5: Top panel displays the intersection between the Freesurfer derived cerebellar gray matter mask and the SUIT-derived inferior cerebellum mask. The lower panel displays the superior cerebellar 'exclusion' mask.

- To minimize partial volume effects, the resulting mask is eroded by one voxel (Figure 6).

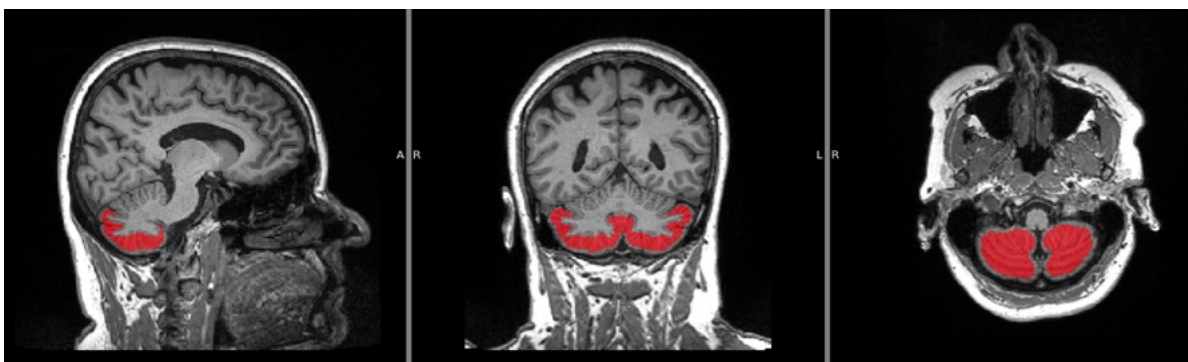


Figure 6: The final inferior cerebellar mask.

ROI Processing

- For the meta-ROI analysis, the Desikan-Killiany atlas is thresholded to extract composite regions corresponding to Braak stages I, III-IV, and V-V. A meta-temporal composite inclusive of the entorhinal, amygdala, fusiform, inferior and middle temporal regions is also computed (Figure 7). The corresponding Freesurfer labels can be found towards the end of this document.

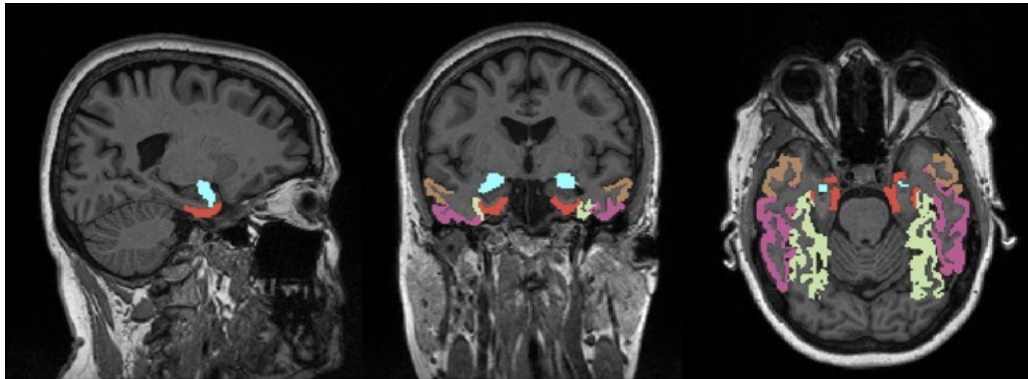


Figure 7: Tau-PET ROI: Meta-Temporal Composite

QC for Reference Region and Meta-ROIs:

- The masks are overlaid on the native T1 image and visually inspected for under-/over-segmentations and encroachment into the meninges.
 - If each mask covers the respective gyri/sulci across all slices, the subject can be passed.
 - If a meta-ROI is grossly under-segmented (e.g underestimation of the entorhinal cortex in Braak I/Meta-temporal mask) we fail that composite region and halt processing for that subject.
- Post extraction, the reference region and ROIs masks are transformed back to PET space using the inverse of the PET-to-T1 transformation matrices (employing the Nearest Neighbor interpolation method) (Figure 8).



Figure 8: Inferior cerebellum overlaid on the T1 image in PET space.

Step 5: Extraction of FTP Standardized Uptake Value Ratios (SUVRs)

- The PET scan is normalized using the inferior cerebellum as the reference region.
- Non-weighted Standardized Uptake Value Ratios (SUVRs) for each ROI are calculated by dividing the mean uptake of the composite by that of the inferior cerebellum.
- Derivation of SUVR statistics for Freesurfer wide regions is done by processing the aparc+aseg file (DK atlas) through mri_segstats.
- To calculate volume-weighted results, the SUVR of each component within a composite region is multiplied by its FreeSurfer-derived volume. The weighted values are then summed and divided by the total volume of the composite region.

FreeSurfer Regions (ADNI)

Braak-I Composite Region

1006 : ctx-lh-entorhinal

2006 : ctx-rh-entorhinal

Braak-III-IV Composite Region

1016 : ctx-lh-parahippocampal

1007 : ctx-lh-fusiform

1013 : ctx-lh-lingual

18 : Left-Amygdala

2016 : ctx-rh-parahippocampal

2007 : ctx-rh-fusiform

2013 : ctx-rh-lingual

54 : Right-Amygdala

1015 : ctx-lh-middletemporal

1002 : ctx-lh-caudalanteriorcingulate

1026 : ctx-lh-rostralanteriorcingulate

1023 : ctx-lh-posteriorcingulate

1010 : ctx-lh-isthmuscingulate

1035 : ctx-lh-insula

1009 : ctx-lh-inferiortemporal

1033 : ctx-lh-temporalpole

2015 : ctx-rh-middletemporal

2002 : ctx-rh-caudalanteriorcingulate

2026 : ctx-rh-rostralanteriorcingulate

2023 : ctx-rh-posteriorcingulate

2010 : ctx-rh-isthmuscingulate

2035 : ctx-rh-insula

2009 : ctx-rh-inferiortemporal

2033 : ctx-rh-temporalpole

Braak-V-VI Composite Region

1028 : ctx-lh-superiorfrontal

1012 : ctx-lh-lateralorbitofrontal

1014 : ctx-lh-medialorbitofrontal

1032 : ctx-lh-frontalpole

1003 : ctx-lh-caudalmiddlefrontal

1027 : ctx-lh-rostralmiddlefrontal

1018 : ctx-lh-parsopercularis
1019 : ctx-lh-parsorbitalis
1020 : ctx-lh-parstriangularis
1011 : ctx-lh-lateraloccipital
1031 : ctx-lh-supramarginal
1008 : ctx-lh-inferiorparietal
1030 : ctx-lh-superiortemporal
1029 : ctx-lh-superiorparietal
1025 : ctx-lh-precuneus
1001 : ctx-lh-bankssts
1034 : ctx-lh-transversetemporal
2028 : ctx-rh-superiorfrontal
2012 : ctx-rh-lateralorbitofrontal
2014 : ctx-rh-medialorbitofrontal
2032 : ctx-rh-frontalpole
2003 : ctx-rh-caudalmiddlefrontal
2027 : ctx-rh-rostralmiddlefrontal
2018 : ctx-rh-parsopercularis
2019 : ctx-rh-parsorbitalis
2020 : ctx-rh-parstriangularis
2011 : ctx-rh-lateraloccipital
2031 : ctx-rh-supramarginal
2008 : ctx-rh-inferiorparietal
2030 : ctx-rh-superiortemporal
2029 : ctx-rh-superiorparietal
2025 : ctx-rh-precuneus
2001 : ctx-rh-bankssts
2034 : ctx-rh-transversetemporal
1021 : ctx-lh-pericalcarine
1022 : ctx-lh-postcentral
1005 : ctx-lh-cuneus
1024 : ctx-lh-precentral
1017 : ctx-lh-paracentral
2021 : ctx-rh-pericalcarine
2022 : ctx-rh-postcentral
2005 : ctx-rh-cuneus
2024 : ctx-rh-precentral
2017 : ctx-rh-paracentral

Meta-Temporal Composite Region

18 : Left-Amygdala
1006 : ctx-lh-entorhinal
1007 : ctx-lh-fusiform
1009 : ctx-lh-inferiortemporal
1015 : ctx-lh-middletemporal
54 : Right-Amygdala
2006 : ctx-rh-entorhinal
2007 : ctx-rh-fusiform
2009 : ctx-rh-inferiortemporal
2015 : ctx-rh-middletemporal

Dataset Information

Dataset Name	Scans Processed (IDA)	Date Submitted
PPG_Tau-PET_AV1451	Accelerated Sagittal MPRAGE (3T) AA-APOE_TAU (AC)	3rd December 2024

References

1. Landau, S., Ward, T. J., Murphy, A., & Jagust, W. Flortaucipir (AV-1451) processing methods.https://adni.bitbucket.io/reference/docs/UCBERKELEYAV1451/UCBERKELEY_AV1451_Methods_2021-01-14.pdf
2. Diedrichsen, J., *A spatially unbiased atlas template of the human cerebellum*. Neuroimage, 2006. 33(1): p. 127-38.
3. Tustison, N.J., Cook, P.A., Holbrook, A.J. et al. The ANTsX ecosystem for quantitative biological and medical imaging. Sci Rep 11, 9068 (2021). <https://doi.org/10.1038/s41598-021-87564-6>